

A2UI v0.9 on Gemini Enterprise

Design Handbook

Building interactive UI natively in Gemini Enterprise chat
Confirmed behaviours · component reference · what's additionally possible

Basic catalog (public) — with notes on the Material catalog

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1. What A2UI v0.9 is

A2UI (Agent-to-UI) is a declarative protocol: an agent sends UI messages and the client renders real, interactive components — buttons, forms, pickers, tabs, modals, images — directly inside the chat surface. No front-end code is deployed; the agent describes the UI as JSON and the renderer draws it.

Gemini Enterprise (GE) renders A2UI **v0.9** (available since 2026-05-13)^{[1][5]}, alongside the older v0.8. v0.9 ships **two catalogs** of components^[2]:

- **Basic catalog** — the portable, framework-agnostic set. Its catalogId is public and it renders in GE today. Full layout + input components, with limited styling. *This handbook is primarily about the basic catalog.*
- **Material catalog** — Material Design 3 components with real styling (per-component colour, elevation, appearance, a true dropdown, table, accordion). It is GE-proprietary; its catalogId is not published. Covered in section 9.

Golden rule: every A2UI v0.9 message must carry "version": "v0.9". Without it, GE falls back to v0.8 parsing and renders nothing.

2. What changed from v0.8 to v0.9

If you built on v0.8, these are the format changes. v0.9 is designed to be flatter and easier for a model to generate.^[3]

Area	v0.8	v0.9
Component shape	Nested key wrapper: { "component": { "Text": { ... } } }	Flat discriminator: { "id": "t", "component": "Text", "text": "... " }
Messages	surfaceUpdate · dataModelUpdate · beginRendering	createSurface · updateComponents · updateDataModel · deleteSurface
Version	Implicit	Explicit "version": "v0.9" on every message (required)
Catalog	Standard only	catalogId on createSurface (basic or material URL)
Data values	Typed wrappers { "literalString": "x" }	Implicit typing — just "x" / 5
Binding	{ "path": "/a/b" }	Unified path (JSON Pointer) everywhere
Formatting	Done in agent code	formatString with \${...} interpolation
Validation	Agent-side checks	Declarative checks (client-side functions)
Styling	usageHint, no colours	variant + (material) colour / appearance / elevation / theme
Button emphasis	primary: true/false	variant: primary borderless (basic)

Several v0.8 practices still apply in v0.9: strip A2UI JSON from model history so it isn't echoed into chat, use a fresh surfaceId per surface, route clicks deterministically in code, and keep server-side validation as a fallback.

3. Message format & the required recipe

A surface is built from a short sequence of messages, each a separate A2A DataPart (mimeType application/json+a2ui), each carrying the version marker:

1. **createSurface** — opens a surface; declares catalogId (and optional theme).
2. **updateComponents** — the flat list of components, referenced by id.
3. **updateDataModel** — seeds/updates bound data (path: "/", value:{...}).

There is no beginRendering in v0.9 — createSurface + updateComponents are enough to render.

The public basic catalogId

```
https://a2ui.org/specification/v0_9/catalogs/basic/catalog.json
```

(This URL is the live catalog schema.^[6])

catalogId must be the catalog's **full URL** (it equals the catalog's \$id); GE looks it up in a registry. A short id like "material" is rejected with "Catalog not found".

Copy-paste recipe

```
{
  "version": "v0.9",
  "createSurface": {
    "surfaceId": "s",
    "catalogId": "https://a2ui.org/specification/v0_9/catalogs/basic/catalog.json",
    "sendDataModel": false
  },
  "version": "v0.9",
  "updateComponents": {
    "surfaceId": "s",
    "components": [
      {
        "id": "root",
        "component": "Column",
        "children": ["title", "btn"],
        "align": "stretch"
      },
      {
        "id": "title",
        "component": "Text",
        "text": "Hello",
        "variant": "h4"
      },
      {
        "id": "btn",
        "component": "Button",
        "child": "lbl",
        "variant": "primary",
        "action": {
          "event": {
            "name": "clicked"
          }
        }
      },
      {
        "id": "lbl",
        "component": "Text",
        "text": "Primary"
      }
    ]
  }
}
```

4. Basic catalog — component reference

The complete basic catalog: Text, Image, Icon, Video, AudioPlayer, Row, Column, List, Card, Tabs, Modal, Divider, Button, TextField, CheckBox, ChoicePicker, Slider, DateTimeInput^[6]. Key shapes:

Component	Key properties	Notes
Text	text, variant (h1–h5 / body / caption)	variant is the only size control. Renders Markdown bold + links.
Button	child (id of a Text = the label), variant (default / primary / borderless), action	Label is a child Text — not a text/label prop.
Column / Row	children (array of ids), justify, align	justify = main axis, align = cross axis. See section 5.
Card	child (a SINGLE id)	Wrap multiple items in a Column/Row and pass that id. Auto padding.

Component	Key properties	Notes
ChoicePicker	options, value (binding), variant (mutuallyExclusive / multipleSelection), displayStyle (chips / checkbox)	No dropdown style. filterable exists in schema but is ignored in GE.
Slider	value (binding), min, max, label	Renders with a value label.
CheckBox	value (binding), label	Boolean toggle.
DateTimeInput	value (binding), enableDate, enableTime	Native date and/or time pickers.
TextField	value (binding), label, variant	Single/short/long text input.
Image	url, fit (contain/cover/fit/...), variant (icon...largeFeature), description	External URLs render in v0.9 (blocked in v0.8).
Tabs	tabs: [{ title, child }]	Each tab points to one child component id.
Modal	trigger (id), content (id)	Opens an overlay on trigger click.
Divider	axis (horizontal / vertical)	Rule line.
Icon	name	Renders nothing if the name isn't in GE's set (degrades gracefully).

5. Layout system

Layout is done with Column and Row. Each takes children (an array of component ids) plus two axis controls:

- **justify** (main axis): start · center · end · spaceBetween · spaceAround · spaceEvenly · stretch
- **align** (cross axis): start · center · end · stretch

Setting **align** on a surface's root Column visibly repositions the whole card in GE: start hugs left, center centres every child, stretch fills the width ("justified"). This is a real, user-selectable global layout lever.

Adding space (there is no padding property)

Basic components have no padding / margin / gap property. Two ways to add breathing room:

- Use **justify**: spaceBetween / spaceAround / spaceEvenly to distribute children.
- Inject spacer components — a Text holding a non-breaking space (\u00A0) between items (a plain trailing space is collapsed by the renderer).

Tight text fix: don't pack many lines into one Text with newlines — split each line into its own Text inside a Column. Separate components get real vertical rhythm; use variant body over caption for size.

Building a table (there is no Table component)

The basic catalog has no Table (MaterialTable is Material-only). Build one from layout primitives:

- **✓ 2-column table (label ↔ value) — reliable.** A Column of Rows, each Row with **justify**: spaceBetween — first cell hugs left, second hugs right, and columns align because every row splits the same width. Add a header Row + a Divider.
- **✗ 3+ column grid — risky.** Needs weight on each cell to line columns up; weight alignment is not confirmed in GE and may render ragged. Use MaterialTable for real multi-column tables.

```
Column[
  Row(justify:spaceBetween)[ Text("Dish",caption),  Text("Price",caption) ],
  Divider,
  Row(justify:spaceBetween)[ Text("Margherita",body), Text("$14.00",body) ],
  ...
]
```

6. Styling — what you can and cannot control

On the basic catalog the styling budget is deliberately small. Know the levers, and know the walls.

You CAN control

- ✓ **Surface theme.primaryColor** (a hex on createSurface.theme) — brands primary buttons, active chip fill, and slider tracks. The one global colour lever. The theme object also takes iconUrl and agentDisplayName.
- ✓ **Text variant** (h1–h5 / body / caption) — this is your font sizing.
- ✓ **Button variant** (default / primary / borderless) and Image variant (icon / avatar / .../ largeFeature).
- ✓ **Layout** via Row/Column align + justify, and weight (proportional size within a Row/Column).
- ✓ **Markdown in Text** — bold and links render (e.g. a map link [Open in Maps](https://...)). HTML and images in Text do not.
- ✓ **ChoicePicker displayStyle** — chips (compact) or checkbox (list).

You CANNOT control (basic catalog)

- ✗ **Per-component padding, margin, font-size, or colour** — only theme.primaryColor globally + component variants. Cards apply padding automatically.
- ✗ **Rounded corners / elevation / shape** on Card or Modal — box shape is renderer-decided.
- ✗ **Background / surface colour** — only primaryColor is honoured; extra theme colour keys are ignored.
- ✗ **A true collapsing dropdown** — ChoicePicker offers only chips / checkbox. filterable (search box) is ignored in GE.
- ✗ **A native Map component** — use a Markdown map link instead.

All of these — real padding, corners, typography, per-component colour, dropdown, background — are **Material catalog** features (section 9).

7. Confirmed behaviours in Gemini Enterprise

Verified by deploying to GE (not just from the spec). The Agent Engine Playground never renders A2UI — it shows raw inlineData; only GE chat renders.

Confirmed working

- ✓ Basic catalog renders end-to-end (Card, Column/Row, Text, Button).
- ✓ Button variant styling: primary = filled, borderless = text, default = outlined.
- ✓ theme.primaryColor repaints primary buttons, active chips, and slider tracks.

- ✓ ChoicePicker (chips) selection, Slider, CheckBox, DateTimeInput all render natively.
- ✓ External images render (the v0.8 block is lifted); multiple images at once are fine.
- ✓ Tabs (title + child per tab) render and switch; Modal opens an overlay on trigger.
- ✓ Markdown bold + links render in a Text with variant body.
- ✓ Data-binding write-back works with flat single-segment paths (/field).
- ✓ action.event.context arrives pre-resolved (real selected values, not raw bindings).
- ✓ align on the root Column repositions the whole surface (Left / Center / Justify).

Confirmed NOT possible / not honoured (basic catalog)

- ✗ Per-component padding, font-size, colour, rounded corners, elevation.
- ✗ Background / surface colour (only primaryColor is honoured).
- ✗ ChoicePicker filterable search box (silently ignored).
- ✗ A true dropdown (only chips / checkbox).
- ✗ Markdown bold inside a caption Text (use body).
- ✗ Changing the "User action triggered." bubble text (see section 8).

8. Interaction & data model

- **Firing events:** a Button carries `action.event = { name, context }`. context values may be literals or {path} bindings.
- **Incoming clicks:** arrive as `{ "action": { name, surfaceId, context, timestamp } }`. Context bindings are pre-resolved by GE, so you read real values — no raw-JSON parsing.
- **Data binding:** bind inputs with `{ "path": "/field" }` and seed via `updateDataModel`. Keep paths **flat and single-segment** — GE writes back only to top-level keys (/email, not /form/email).

"User action triggered." bubble: GE shows this fixed system bubble for every click. The v0.9 Action schema is closed (name + context only) — there is no field to change it. Mitigation: own the agent's own reply text and prepend a readable echo of what was tapped.

9. What the Material catalog additionally unlocks

The Material catalog is Material Design 3 with real styling knobs. It directly fixes the basic-catalog walls above.^[4]

Capability	Material component(s)
True dropdown / select	MaterialSelect
Per-component colour & appearance	color (primary/accent/warn); button appearance (text/filled/elevated/outlined/tonal)
Data table	MaterialTable
Collapsible / accordion	MaterialExpansionPanel
Even grid layout	MaterialGridList, MaterialChips

Capability	Material component(s)
Loading state	MaterialProgressBar, MaterialProgressSpinner
Rounded / elevated cards & dialogs	MaterialCard (raised/outlined), MaterialDialog
Rich inputs	MaterialInput, MaterialDatepicker, MaterialTimepicker, MaterialRadioButton, MaterialSlideToggle

Obtaining the Material catalogId

The Material catalogId is a **negotiation token**, not a fetchable URL — GE's renderer holds a private registry, so the string is GE-internal and not published on the spec site or in open source^{[7][8]}. To obtain it (requires GE/GCP access):

1. **DevTools network capture** — open a GE experience that renders Material, then read the catalogId inside its createSurface in the streamed response.
2. **Cloud Logging** — log the incoming A2A request/metadata from your own agent; GE may advertise supported catalogs during negotiation.
3. **Agent Garden sample / GE support** — a GE-provided A2UI sample's source, or the Admin console.

Once found, set it as catalogId and switch component names (Button → MaterialButton, ChoicePicker → MaterialSelect, ...). Note: a single surface renders against **one** catalog/version — you cannot mix basic and Material (or v0.8 and v0.9) components in the same surface.

10. Build checklist

- **Version marker:** Put "version":"v0.9" on every message — the #1 cause of "renders nothing".
- **Full-URL catalogId:** Use the complete catalog URL, not a short name.
- **Fresh surfaceld:** New surfaceld per surface, unless deliberately updating one in place.
- **Flat data paths:** Bind to single-segment paths (/field) for write-back.
- **Markdown = body:** Put Markdown in a variant:body Text, never caption.
- **Scrub history:** Remove A2UI JSON from model history so it isn't echoed into chat.
- **Deterministic routing:** Route clicks by action.name in code; don't trust model markers.
- **Validate server-side:** Keep a server-side validation fallback even if using declarative checks.
- **Echo the click:** Prepend a readable echo in your reply to soften the fixed system bubble.

References

Inline markers [n] above link to these sources. All links accessed 2026-09; Gemini Enterprise docs and the A2UI spec are updated periodically.

- [1] A2UI v0.9 announcement — Google Developers Blog — <https://developers.googleblog.com/a2ui-v0-9-generative-ui/>
- [2] A2UI v0.9 specification — <https://a2ui.org/specification/v0.9-a2ui/>
- [3] A2UI v0.8 → v0.9 evolution guide — <https://a2ui.org/specification/v0.9-evolution-guide/>

- [4] Gemini Enterprise — A2UI component gallery reference — <https://docs.cloud.google.com/gemini/enterprise/docs/a2ui-agents/a2ui-component-gallery-reference>
- [5] Gemini Enterprise — release notes — <https://docs.cloud.google.com/gemini/enterprise/docs/release-notes>
- [6] A2UI basic catalog (schema JSON) — https://a2ui.org/specification/v0_9/catalogs/basic/catalog.json
- [7] Guide to Gemini Enterprise and A2UI integration — Google Cloud Blog — <https://cloud.google.com/blog/topics/developers-practitioners/guide-to-gemini-enterprise-and-a2ui-integration>
- [8] A2UI — Catalogs (concepts) — <https://a2ui.org/concepts/catalogs/>